



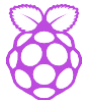
Retro Pi Gaming Project

Welcome! We're excited to share our Retro Pi gaming project. This project combines retro gaming with DIY electronics to create a custom gaming device.

It will feature a Raspberry Pi and a joystick.



Project Components



Raspberry Pi

The brain of our gaming device. It emulates classic consoles.



Joystick

Provides authentic retro gaming control, enhancing the experience.



SD Card

Stores the OS and game ROMs, allowing for game storage.

Project Motivation

Nostalgia

Relive classic gaming experiences from our childhoods.

DIY Fun

Enjoy building our own gaming device while learning about electronics.

Customization

Create a unique gaming experience tailored to our preferences.





Retro Pi Features

Emulation

Play thousands of classic games from multiple consoles.

Customization

Customize the interface and gaming experience to your liking.

Portability

Take your retro games anywhere you go with ease.

Research and Planning

1

Week 1: Research

In the first week, we focused on comprehensive research. We started by learning about RetroPie, Raspberry Pi, Arduino, and the DHT22 sensor. We planned how to connect all components and looked into how emulators and ROMs work.

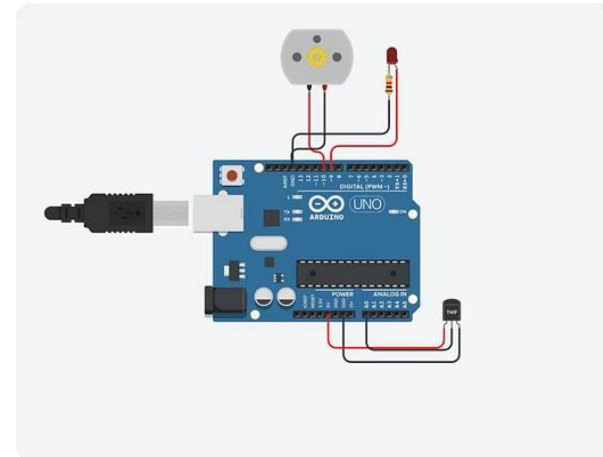


2

Week 2: Setup and coding

In week 2 , we started working on code for system to work. We coded scripts to integrate the Arduino and DHT22 sensor with the system.

```
1 #include "DHT.h"
2
3 #define DHTPIN 2
4 #define DHTTYPE DHT22
5 #define RED_LED 9
6 #define Fan 10
7
8 DHT dht(DHTPIN, DHTTYPE);
9
10 void setup() {
11   Serial.begin(9600);
12   dht.begin();
13
14   pinMode(RED_LED, OUTPUT);
15   pinMode(Fan, OUTPUT);
16
17   Serial.println("Automatic Temperature Control System");
18 }
19
20 void loop() {
21   float temperature = dht.readTemperature();
22   int threshold = 25;
23   Serial.print("Temperature: ");
24   Serial.print(temperature);
25
26   if(temperature > 20){
27     digitalWrite(RED_LED, HIGH);
28   }
29   if (temperature > threshold) {
30     digitalWrite(Fan, HIGH);
31     digitalWrite(RED_LED, HIGH);
32     Serial.println("Cooling ON");
33   } else {
34     digitalWrite(Fan, LOW);
35     digitalWrite(RED_LED, LOW);
36     Serial.println("Cooling OFF");
37   }
38
39   delay(2000);
40 }
```





Completed Tasks

1

Integrated Joystick Library

Integrated joystick library with the Retro Pi setup.

2

Testing

Verified library functionality with sample code.

Next Steps: Code and Assembly

1

Code Development

Write the core gaming logic and integrate joystick inputs.

2

Hardware Assembly

Connect joystick to Raspberry Pi and assemble components.

3

Testing

Test game functionality and ensure smooth gameplay.

Project Takeaways

Retro Gaming is Fun

It is also educational and brings back nostalgic memories.

DIY Electronics

Rewarding and allows for extensive customization options.

Classic Gaming Experience

Our project will deliver this on a custom device.



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